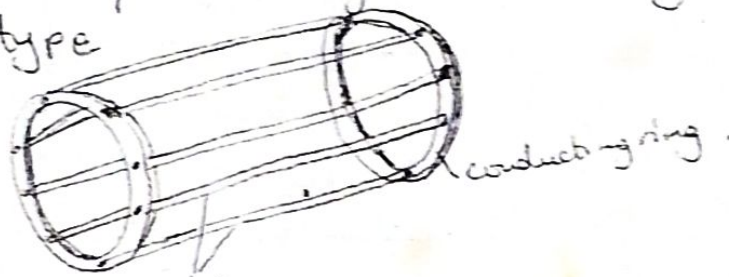


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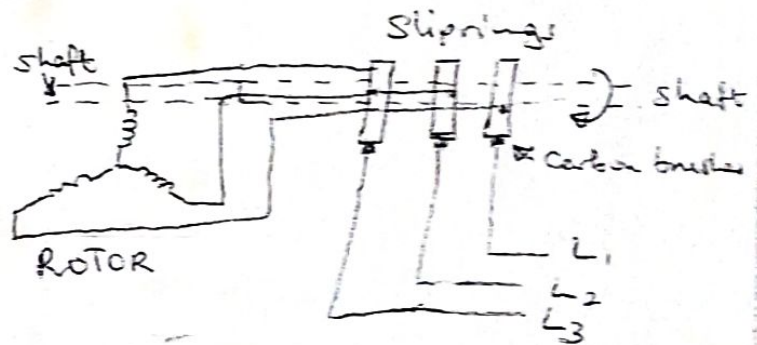
THREE PHASE MOTORS TA

Basically a 3-phase machine has three sets of stator coils uniformly distributed around the stator core, and connected in either star or delta.

The rotor used is either Squirrel Cage or wound type - in which case the motor is either called a Squirrel cage induction type or a wound ~~ind~~ rotor type.



rotor conductors
Cage rotor.



WOUND ROTOR

THREE-PHASE INDUCTION MOTORS

This motor is the simplest and cheapest of all motors and requires least maintenance because it has no commutator.

The starting torque is high and the starting current is also high. For this reason, except for very small motors, a starting equipment is necessary to make the motor start with a reduced voltage and a reduced torque.

Again an induction motor is either of the squirrel cage or wound rotor type.

(a) Squirrel cage 3-phase induction motor.

This motor has a wound 3-phase stator and a

(2)

Squirrel-Cage rotor. There are no slip-rings, and (unlike with the wound rotor type) and thus no moving connections. For this reason the motor is:

- robust
- simple, and
- cheap.

A Squirrel cage rotor consist of single armature conductors laid in insulated slots in the rotor surface. At each end of the rotor is a short-circuiting ring to which each armature conductor is connected. This gives the appearance of a circular cage, hence the name.

For better running and starting the slots on the rotor are skewed. Sometimes aluminium conductors are used. Since there are no external connections this type of rotor eliminates the use of slip-rings and brushes.

Torque & Speed characteristic

The speed of a Squirrel cage 3 phase motor is similar to that of a shunt motor, and operate slightly below the synchronous speed, (about 96-97 percent). The remaining 3% is called the "slip". For bigger machines the slip may drop to as low as 1.5% while it may increase to about 4% for smaller machines.

The starting torque is smaller, and the motor is then only suitable for light duties, where speed control is not essential.

This machine has low power factor at start, but improves at full load i.e. from about 0.85 to 0.9 (lagging).

(3)

Operation of a squirrel cage 3-phase motor.

When a 3phase supply is switched on to the stator winding, a rotating field is set up. This induces an e.m.f in the rotor and so currents flow in these windings.

At starting ~~the~~ when the rotor is stationary, the currents are of supply frequency. The rotor turns in the same direction as the field until it reaches nearly synchronous speed i.e. with only a small slip. The currents are now of slip frequency and of comparatively low values.

If the rotor were to run at synchronous speed, no currents would flow in the rotor, and there would be no torque to turn the rotor.

When the motor is mechanically loaded, the rotor slows slightly, the frequency and the values of the rotor currents increase, and more power is taken by the stator from the mains.

Starting

For all except the smallest motors, it is necessary to reduce the starting current. With the ordinary squirrel cage motor this is achieved by reducing the voltage supplied to the stator.

The torque is proportional to the square of the voltage and so is small at the reduced voltage. Line resistance starting is sometimes used but it is inefficient.

NB If the motor is started by direct-on switch ^{510 12 3 phase 9} the current is large and of low power factor. So the following starting methods are employed:

- Direct-on-line where starting surge is permitted.
- Star-delta starting for motors up to about 20 Kw
- Auto-transformer starting for motors between 20 and 250 Kw.

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Application of a squirrel cage 3-phase motor. (used in conditions of off-load or very light load starting)

— used for general purposes where almost a constant speed is required, and load can be applied after starting. ~~These includes light load starting~~ such as machine drives.

WOUND ROTOR 3-PHASE INDUCTION MOTOR.

The stator winding is wound 3-phase as in the case of the squirrel-cage motor. The rotor usually consists of a slotted armature with ~~connection~~ can equally a 3-phase star connected winding. Mesh connection can equally be used although it is less common. The ends of the phases are brought out to three slip-rings mounted on the rotor shaft. For this reason this motor is sometimes known as the "slipring" motor. Brushes bearing on the rings connect the phases to a tapped 3-phase resistance.

To start the motor, the full mains voltage is switched on to the stator by means of a 3-pole switch and fuses. The rotor resistances are fully in the circuit, thus cutting down the induced currents in the rotor and limiting the supply current to the stator. The rotor begins to rotate and the rotor resistances are cut out - step by step and completely. The sliprings as the motor gains its full speed. The sliprings are short circuited

and the brushes are lifted from the rings by the use of a hand operated device. This method gives a good starting torque and enables the motor to start against load without an unreasonable heavy starting currents.

Starting method and Speed/torque control

The rotor starting resistance may be used for speed control purposes down to about 50% of the synchronous speed. In this case, the resistances must be of the heavy duty type, able to withstand continuous use. This method of speed control is inefficient owing to the loss of energy in the resistances. However the speed/torque performance can be varied by the addition of resistance in the rotor circuit via the sliprings.

Essentially a wound rotor 3-phase motor has no speed control. The use of a rotor resistance controller is wasteful, lowers efficiency, and adds to the cost.

The type of starter employed is the "rotor resistance starter" as shown below.

Rotor Starting of ^{wound} 3-phase Induction motor

Application

A wound rotor 3ph induction motor is used in the following for general purposes requiring almost constant speeds and on ^{no}-load starting.

(6)

These include
~~and~~

- large lathe drives
- pumps (ram type)
- Sawmill machines
- air compressor
- crushing mills
- Cranes
- hoists etc.

Reversal of the direction of rotation

Any 3-phase motor may have its direction of rotation reversed by interchanging any two of the three line connections to the motor.

⑦
3PHASE
STARTING METHODS FOR INDUCTION MOTOR

There are three items to be ^{known} considered when starting an induction motor:

- (i) Torque is determined by rotor resistance and may be small ~~hence~~ the motor should be started on no-load or with very light load
- (ii) Starting current may be excessive if the rotor is short-circuited. This is particularly important for the cage motor.
- (iii) Reduced voltage starting will seriously reduce torque.
→ i.e. line resistance ~~at~~ reduces the starting voltage - hence the torque

Owing to the above reasons, all except the smallest motors require a reduction of current at starting. With the squirrel-cage motor this is done by reducing the voltage supplied to the stator. The torque is proportional to the square of the voltage and ^{therefore it is} ~~is~~ small at the reduced voltage (i.e. $T \propto V^2$).
Line resistance starting is sometimes used but it is not sufficient.

NB If the motor is started by direct on switch, the current is large and of low power factor.

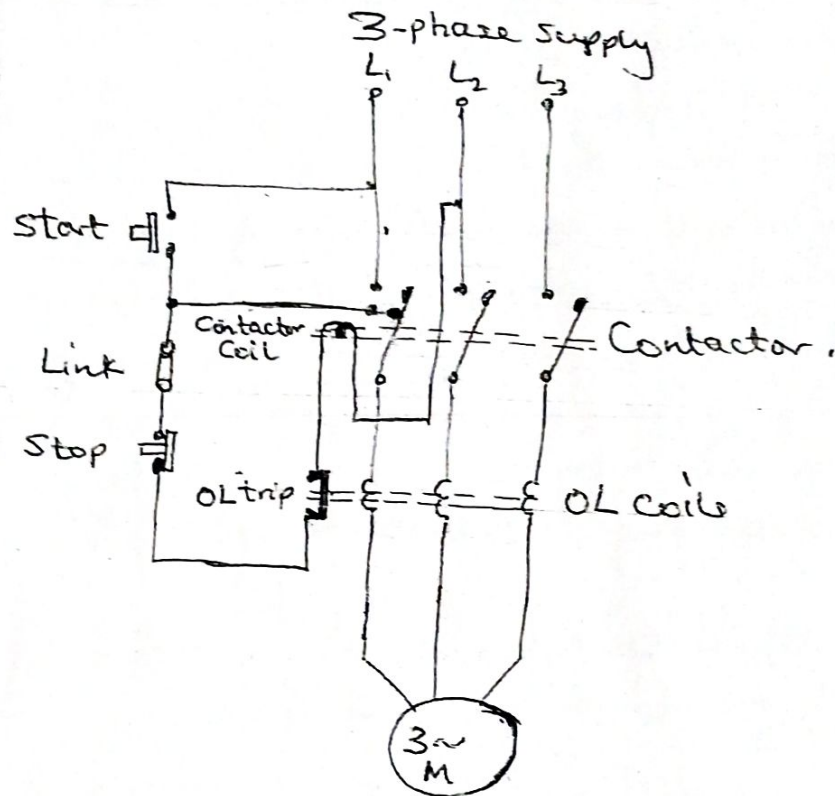
The following three methods are to start the cage induction motor & (unless specially constructed (with a high resistance rotor) starting should be on no-load or very light load.

(i) Direct-on-line starting

As the name implies the supply is directly switched on to the motor through a suitable contactor starter. The ~~supply~~ starting current is high and may cause interference with supplies to other consumers. For this reason the supply authority generally restricts the use of this method to small

⑧

motors of say not more than 2 kw (about $2\frac{1}{2}$ - 3 hp). However, if the supply ~~to such~~ is isolated as may be the case in big factories, there is no such limit and ~~to~~ motors of several hundred kilowatts may be started by the direct-on-line method. The ~~figure~~ ^{diagram} below shows the basic connection of this starting method.



Contactor starter for direct-on-line starting.

Operation

When the ~~Start~~ START button is pushed the magnetic operating coil is energised from two of the line conductors, and four sets of contacts close, providing current to the motor - and also short-circuiting the start button. The operating coil holds the contacts in place while the motor is running and thus the motor ^{can} be stopped by pressing the stop button. This action breaks

the operating circuit, de-energizes the operating coil, breaking the contacts and stopping the motor.

The Overload coils, which are either thermal or magnetic, are so arranged that in times of undue overload they will open a switch in the operating circuit and stop the motor.

Advantages of a D.O.L starter.

- ~~It~~ - provides a quick starting at maximum torque.
- It is reliable, simple and cheap.

Disadvantages.

- It takes a heavy current from the supply system.

Applications.

For general applications which includes:

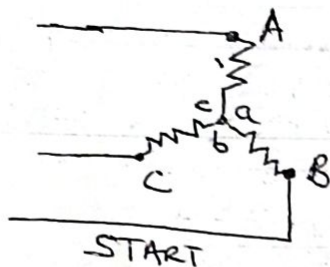
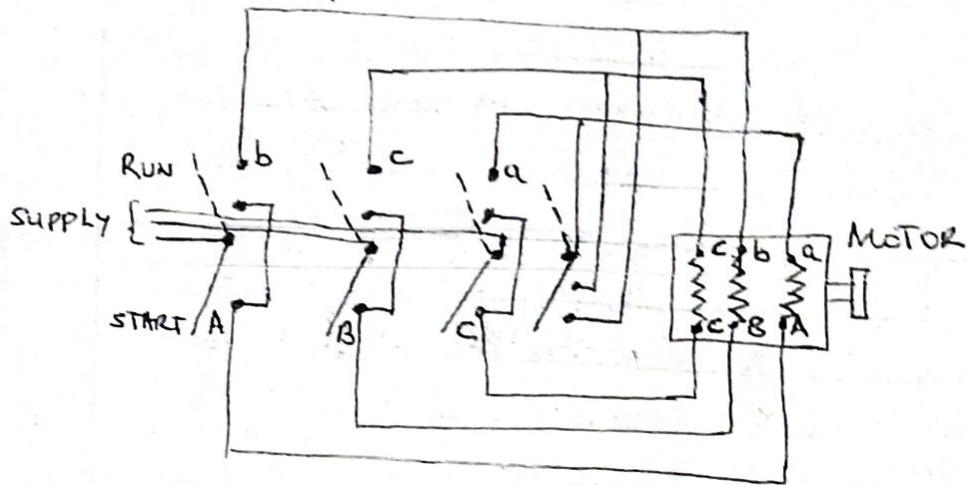
- machines for most small drives
- pumps
- fans
- tools
- small machines

② STAR-DELTA STARTING.

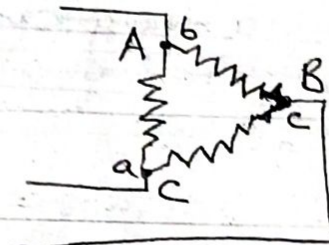
With this method (which is cheap and ^{simple} ~~cheap~~) the six connections to the three stator phase windings of the motor are brought out to the starter. For normal running, the windings are connected in delta, but for starting they are star connected so that the voltage across each phase winding will be $1/1.73$ (i.e. $1/\sqrt{3}$) of the line voltage at the start. Thus the starting voltage per phase is reduced to 57.7% of the running voltage. The changeover from star starting to delta running is made

(10)

by a double-throw switch with interlocks to prevent starting with the switch in the run position.



START



RUN

STAR-DELTA STARTING

A further method of reducing the voltage for starting is by using three phase variable or tapped resistance in the starter circuit in a way similar to the auto-transformer. This method is suitable for small motors, but it is rarely ~~often~~ used, as the other methods give a better starting torque for the same starting current.

NB The change-over can be done by hand, or automatically by a timing device.

Advantages

- this method is cheap & simple.
- It gives a reduced starting current.

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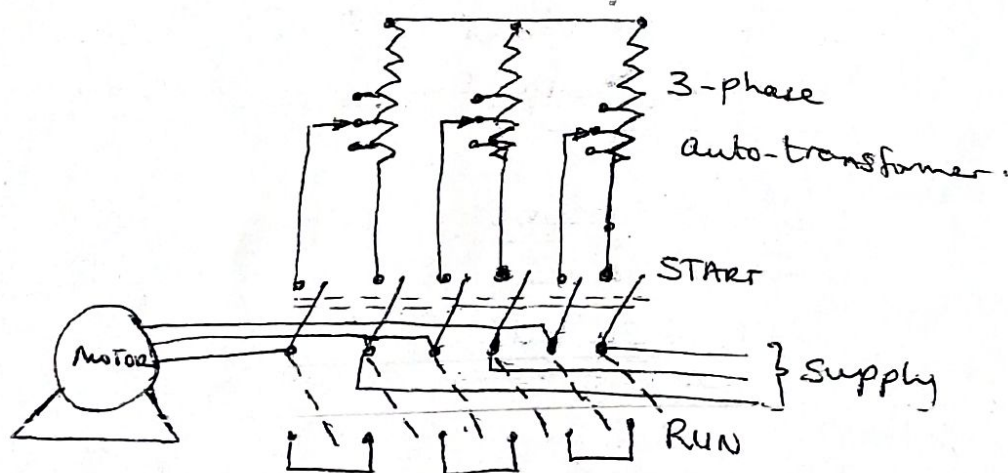
Disadvantages:

- more expensive than D.O.L
- Can only be used for 3-phase machine
- Reduced torque (at starting)
- All ends ~~must~~ of the motor windings must be brought out to the terminal block (box) of the motor

Applications

- line shaft drives
- fans of the centrifugal type etc.
- pump drivers

AUTO-TRANSFORMER STARTING.



A three phase auto-Tx is used for this method in order to reduce the stator voltage, thereby reducing the starting current. This method seriously reduces the starting torque and therefore the voltage is reduced only sufficiently to give the required reduction of the starting current.

Although the auto-Tx may have to provide a ...

(12)

and 80% of line voltage, the starter will be connected permanently to the one set of tappings found most suitable for the particular situation. A double-throw switch connect the auto-Tx in circuit for starting, and after the motor runs up to speed the switch is moved into the run position which connect the supply directly into the motor. Some form of interlocking mechanism is used to safeguard against the switch being closed in the run position while the motor is stopped.

Advantages.

- used where star-delta starter is not applicable i.e. only 3 ends of motor wdg are available.
- Adaptable to operations i.e. starting current and torque can be varied by correct tapping
- Can be used with single phase motor.

disadvantages.

- expensive
- not suitable for starting motors against heavy loads.

Applications

- motor generator sets
- centrifugal machines.
- mixers
- heavy fans.